

Project Work Plan

Course Name: Software Development I

Course Code: CSE 2216

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Project Title: Ping Pong Game

Course: Software Development Lab

Programming Language: Java (Swing)

10-Week Work Plan for Ping Pong Game Project

Week	Phase
1	Project Planning & Concept Finalization
2	Requirements Analysis & System Design
3	Environment Setup & Project Initialization
4	UI Layout & Game Window Development
5	Paddle Movement & Controls
6	Ball Physics & Collision Detection
7	Scoring System & Win Condition
8	AI Opponent & UX Polish
9	Error Handling & Testing
10	Final Testing, Documentation & Submission

Week 1: Project Planning & Concept Finalization

- Select project topic and objectives.
- Define gameplay mechanics, controls, and project scope.

Prepare initial project plan

Week 2: Requirements Analysis & System Design

- Gather functional requirements for paddle movement, scoring, and controls.
- Design system architecture and file layout.
- Create basic wireframes for the canvas and score display.

Week 3: Environment Setup & Project Initialization

- Install JDK and configure the IDE.
- Configure the Java Swing environment.
- Create project structure and main game window class.

Week 4: UI Layout & Game Window Development

- Develop the main game window and play area panel.
- Create the score display and status area.
- Apply theme and basic styling to the GUI.

Week 5: Paddle Movement & Controls

- Implement Player 1 controls (W / S keys) using KeyListener.
- Implement Player 2 / AI controls (Up Arrow / Down Arrow keys).
- Add boundary constraints so paddles stay within the play area.

Week 6: Ball Physics & Collision Detection

- Implement ball movement using the Timer class for the game loop.
- Add collision detection between the ball and paddles.
- Add wall collision so the ball bounces off the top and bottom edges.

Week 7: Scoring System & Win Condition

- Implement real-time score tracking for both players.
- Detect when the ball passes a paddle and update the score.
- Implement the win condition once a player reaches the score limit.

Week 8: AI Opponent & UX Polish

- Implement basic AI paddle behavior for single-player mode.
- Polish the UI with a clean, modern aesthetic and visual feedback.
- Optimize the Timer interval for smooth, fluid gameplay.

Week 9: Error Handling & Testing

- Add exception handling for invalid states and edge cases.
- Test paddle and ball collision across different speeds.
- Fix bugs and optimize overall game performance.

Week 10: Final Testing, Documentation & Submission

- Perform complete gameplay testing.
- Prepare project documentation and screenshots.
- Finalize and submit the project.